

SOFTWARE TEST PLAN (STP)

EV Charging Station Reservation System

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Problem Statement: #21 - Smart Cities, Transport & Logistics

Testing strategy, scope, test approach, resources, and acceptance criteria

1. Purpose

This Software Test Plan defines how the EV Charging Station Reservation System will be verified against its functional and non-functional requirements. The focus is on compatible charger discovery, conflict-free slot reservation, dynamic pricing, booking lifecycle handling, and secure and responsive operation.

2. Scope

In scope: EV driver and station operator workflows; connector compatibility (CCS, CHAdeMO, Type 2); charging-station search; 30-minute reservation; concurrent booking prevention; dynamic power-rate calculation; booking cancellation/release/penalty behavior; validation, access control, and performance under peak load.

Out of scope: physical charger hardware validation, vehicle battery health, electricity-grid control, payment-gateway certification, and field testing of actual charging equipment.

3. Test Objectives

The tests shall demonstrate that each stated requirement behaves as intended, invalid or conflicting reservations are rejected, compatible chargers are presented correctly, expired/unused locked slots are handled correctly, and the system remains secure and responsive under representative load.

4. Test Strategy

Test level	Approach	Purpose
Unit testing	Automated tests for validation, pricing, connector matching, reservation rules, and penalty logic	Verify individual software components
Integration testing	Exercise interactions among driver UI, reservation service, station data, pricing, and persistence	Verify data flow and service integration
System testing	End-to-end scenarios using representative drivers, chargers, slots, and station operator actions	Validate complete system behavior
Security testing	Authentication/authorization checks, input validation, session handling, and unauthorized-access cases	Verify NFR security expectations
Performance testing	Simulated peak reservation/search traffic and response-time measurement	Verify responsiveness and stability
Usability testing	Representative driver workflows for search, compatibility filtering, booking, and cancellation	Verify understandable user interaction

5. Test Environment and Resources

Item	Planned setup
Application	Web-based EV charging reservation application under development
Database	Project persistence layer containing stations, chargers, slots, users, and reservations
Browsers	At least two modern browsers; verify responsive behavior where applicable
Test data	Stations with CCS, CHAdeMO and Type 2 sockets; available, occupied, locked, and expired slots
Load simulation	Concurrent virtual users for search and reservation requests

6. Requirements-Based Test Matrix

Req ID	Test objective	Representative test	Expected result
FR-001	Connector compatibility + reservation	Search using each connector type and reserve an available compatible socket	Only compatible chargers are shown and selected slot is reserved
FR-002	Station/slot discovery	Search by location and availability	Relevant available stations and slots are returned
FR-003	Concurrent booking protection	Two users attempt the same socket/slot simultaneously	Only one reservation succeeds; conflict is rejected

Req ID	Test objective	Representative test	Expected result
FR-004	Dynamic pricing	Calculate price across different demand/power conditions	System applies the configured dynamic rate consistently
FR-005	Reservation lifecycle	Cancel, complete, or miss a booking window	State changes correctly and slot becomes available when rules permit
NFR-001	Performance + security	Run peak-load search/reservation tests and unauthorized access attempts	Latency/security acceptance targets are satisfied; unauthorized actions are blocked
NFR-002	Reliability/usability	Repeat representative reservation and recovery scenarios	No silent data loss; user receives clear status/error feedback

7. Entry and Exit Criteria

Entry: approved requirements are available; test data is prepared; test build is deployable; required services are accessible.

Exit: all planned critical/high-priority tests are executed; no unresolved critical defects remain; failed cases are documented with disposition; requirements have traceable verification evidence.

8. Defect Handling

Each failed test shall be recorded with requirement ID, environment, steps to reproduce, expected result, actual result, severity, evidence, and retest status. Critical defects block release until resolved or formally accepted.

9. Deliverables

Test cases, test data, execution results, defect log, retest evidence, and final test summary/report shall be maintained with the project documentation.